

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date: 29 September 2026

Team ID: SWTID-2026-7735

Project Name: AI Blog Nest API

College: C.T.T.E College for Women

Team Leader: Natshitha.P

Team Members: Nivedha.A, Sanmathi.B, Merlin.Y

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Project Architecture: AI Blog Nest API – secure AI-powered blog management backend

Architecture focus: secure API layer, MongoDB data storage, external LLM integration and cloud deployment.

Table-1 : Components & Technologies:

Table-2: Application Characteristics:

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>

Team Leader: Natshitha.P

Team Members: Nivedha.A, Sanmathi.B, Merlin.Y

Architecture Flow: Client/Web UI → Authentication & Role Guard → REST API → Blog/Comment/Category Services → MongoDB. AI requests are routed to an external LLM service for blog generation and summarization; media assets are stored separately.

Implementation note: The supplied project brief defines the AI Blog Nest API capabilities but does not specify exact vendors/frameworks. The technologies above are therefore a proposed implementation stack aligned with those capabilities.

Date	29 September 2026		
Team ID	SWTID-2026-7735		
Project Name	AI Blog Nest API		
Maximum Marks	4 Marks		
S.No	Component	Description	Technology
1	User Interface	Web interface for readers, authors and administrators.	React.js, HTML5, CSS3, JavaScript

S.No	Component	Description	Technology
2	Application Logic-1	REST API, authentication, authorization, routing and request processing.	Node.js, Express.js
3	Application Logic-2	Blog CRUD engine for content, status, categories/tags and asset references.	Express.js, Mongoose
4	Application Logic-3	Context-aware AI generation and summarization pipeline.	LLM API, prompt templates, Node.js
5	Database	Users, roles, blogs, tags, categories, comments, moderation states and metadata.	MongoDB, Mongoose
6	Cloud Database	Managed cloud database for scalable application storage.	MongoDB Atlas
7	File Storage	Separate storage for blog images and media assets.	Cloudinary / object storage
8	External API-1	External AI service used for blog generation and summarization.	LLM API
9	External API-2	Email/notification service for application notifications where required.	SMTP / transactional email API
10	Machine Learning Model	Generates structured blog posts and concise summaries from supplied context.	Large Language Model (LLM)
11	Infrastructure (Server / Cloud)	Cloud-ready API deployment with managed database and container support.	Node.js runtime, Docker, AWS/Render-style cloud hosting

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Open-source backend, API and database libraries are used for maintainability.	Node.js, Express.js, Mongoose
2	Security Implementations	JWT authentication, role-based route guarding, password hashing, validation, sanitization, security headers, CORS and rate limiting.	JWT, bcrypt, Helmet, express-validator, express-mongo-sanitize

S.No	Characteristics	Description	Technology
3	Scalable Architecture	Layered separation of client, API/business logic, AI integration and data storage allows independent scaling.	3-tier/layered architecture, REST API, Docker
4	Availability	Stateless API instances and managed database services support restart/recovery and cloud availability.	MongoDB Atlas, cloud hosting, health checks
5	Performance	Indexed tag/category queries, pagination, bounded payloads and separated media storage reduce API/database workload.	MongoDB indexes, pagination, caching-ready API design, object storage